

Time Series Analysis of Stock Prices

Codveda Technology – Data Analytics Internship

Submitted By

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Objective

The objective of this project is to analyze stock price data over time, identify trends, calculate moving averages, and perform seasonal decomposition.

Dataset

Dataset Name: Stock Prices Dataset

The dataset contains historical stock price information, including opening price, closing price, highest price, lowest price, trading volume, and date.

For this project, **Apple (AAPL)** stock data was selected for analysis.

Tools Used

- Python
 - Google Colab
 - Pandas
 - Matplotlib
 - Statsmodels
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Methodology

Step 1 – Data Loading

The stock dataset was loaded into a Pandas DataFrame.

Step 2 – Date Conversion

The date column was converted into datetime format and set as the index.

Step 3 – Company Selection

Apple (AAPL) stock data was selected for analysis.

Step 4 – Trend Visualization

The closing stock price was plotted to observe long-term trends.

Step 5 – Moving Average

A 7-day moving average was calculated to smooth short-term price fluctuations.

Step 6 – Seasonal Decomposition

The time series was decomposed into trend, seasonal, and residual components using the Statsmodels library.

Results

- The stock price trend was successfully visualized.
 - The 7-day moving average smoothed daily fluctuations.
 - Seasonal decomposition separated the data into trend, seasonal, and residual components.
 - The analysis provided better insight into stock price behavior over time.
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Conclusion

Time series analysis is an effective technique for understanding stock market behavior. Trend analysis, moving averages, and seasonal decomposition provide valuable insights that can support forecasting and financial decision-making.